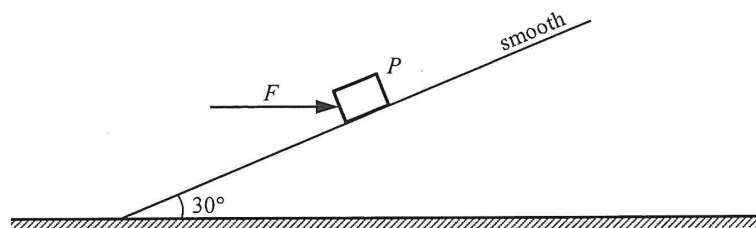


5.

Figure 5.1



- (a) A block P of mass 10 kg is kept stationary on a smooth incline by a horizontal force F as shown in Figure 5.1. The incline makes an angle of 30° with the horizontal. ($g = 9.81\text{ m s}^{-2}$)

- (i) On Figure 5.1, indicate and label all other forces acting on P . (2 marks)
- (ii) Find the magnitudes of the force F and the force exerted by the block on the incline respectively. (3 marks)

Answers written in the margins will not be marked.

- (b) Now F is removed and neglect air resistance.

- (i) What is the magnitude of the acceleration of the block? (1 mark)

- (ii) Explain whether the force exerted by the block on the incline would increase, decrease or remain unchanged when compared with that in (a)(ii). (2 marks)

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- (d) In order to reduce the error contributed by the polystyrene cup, another student suggests repeating the measurements using a copper cup of similar shape and size. Explain whether the suggestion is justified. (2 marks)

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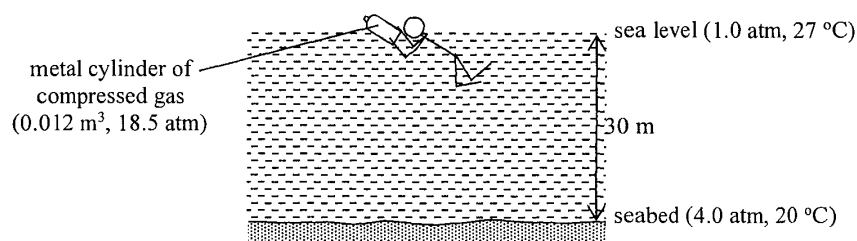
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2. A diver makes a sound by tapping a metal cylinder at sea level. Within a time of 0.04 s, the sound goes vertically to the seabed 30 m below and echoes back to the sea level.

(a) Estimate the speed of sound in sea water.

(2 marks)

Figure 2.1



The metal cylinder of volume 0.012 m³ contains compressed gas under a pressure of 18.5 atm is initially at sea level, where the pressure is 1.0 atm and the temperature is 27 °C. The diver then brings the cylinder to the seabed where the pressure is 4.0 atm and the temperature is 20 °C. Assume that the volume of the cylinder remains unchanged. Given: atmospheric pressure 1.0 atm = 1.01×10^5 Pa

*(b)(i) Show that at the seabed the pressure in the cylinder becomes 18.1 atm.

(1 mark)

(ii) Explain the pressure drop in the cylinder using the kinetic theory.

(2 marks)

Answers written in the margins will not be marked.

- (b) (i) Use the ray diagram drawn to estimate the ratio $\frac{\text{height of object } AB}{\text{distance of } AB \text{ from } L}$. The horizontal and vertical scales are 1:20 and 1:1 respectively. (2 marks)

- (ii) Hence, estimate the height of object AB which is a lamp post at a distance of 200 m from lens L . (1 mark)

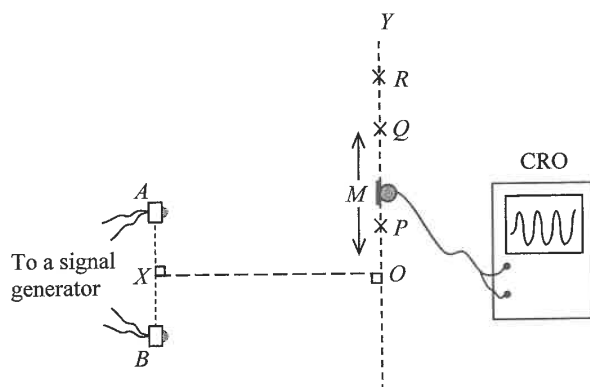
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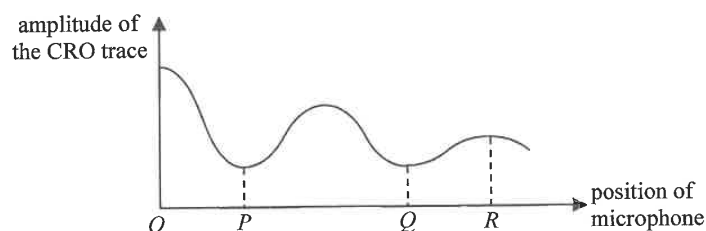
6.

Figure 6.1



In Figure 6.1, two small identical loudspeakers *A* and *B* produce coherent sound waves. *X* is the mid-point of *AB*. A microphone *M* connected to a CRO is moved along *OY* to detect the loudness of the sound, with CRO trace of a larger amplitude representing a greater loudness. Figure 6.2 shows the result.

Figure 6.2



(a) Explain what is meant by **coherent** sound waves.

(1 mark)

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(b) (i) Explain why sound of alternate maximum and minimum loudness is detected along *OY*.

(2 marks)

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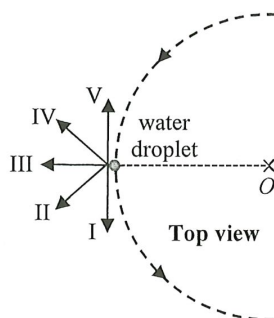
- (ii) Estimate the centripetal force acting on the water droplet shown in Figure 4.2. The mass of the droplet is 2×10^{-5} kg. (2 marks)

Suppose that the water droplet is about to escape via a small hole on the wall of the drum,

- (iii) estimate the speed of the water droplet.

(2 marks)

- (iv) along which direction (I – V) would the droplet travel just as it escapes from the drum ? (1 mark)



Answers written in the margins will not be marked.

5. A sound wave from a loudspeaker is propagating to the right in air. Figure 5.1(a) shows the equilibrium positions of some particles (*a* to *u*) in air. Figure 5.1(b) shows their positions at time $t = 0$.

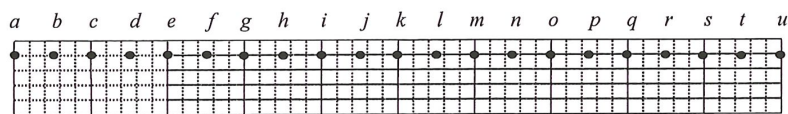


Figure 5.1(a)

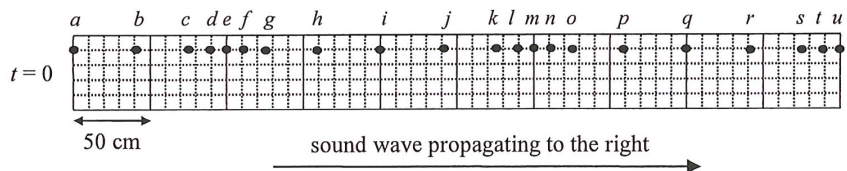


Figure 5.1(b)

- (a) At time $t = 0$, state a particle which

(i) is at a centre of rarefaction.

(1 mark)

(ii) has the greatest displacement to the right from its equilibrium position.

(1 mark)

- (b) The width of each grid is 50 cm as shown in Figure 5.1(b). The speed of sound in air is 340 m s^{-1} .

(i) What is the wavelength of the sound ?

(1 mark)

(ii) Find the frequency of the sound.

(2 marks)

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- *6. After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation d .

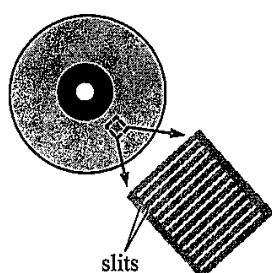


Figure 6.1(a)

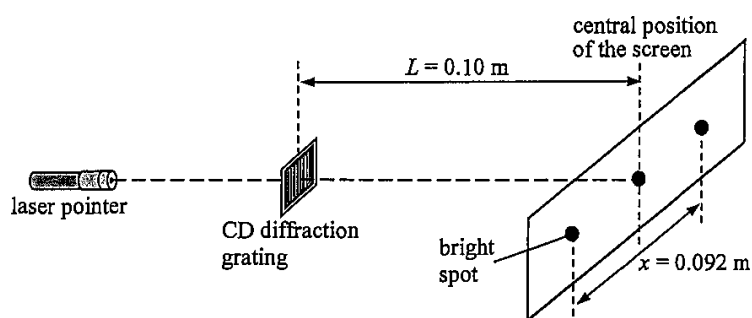


Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen, L , is 0.10 m, the separation between the first-order maxima, x , is 0.092 m and the wavelength λ of the red laser light beam used is 650 nm.

(a) Estimate d .

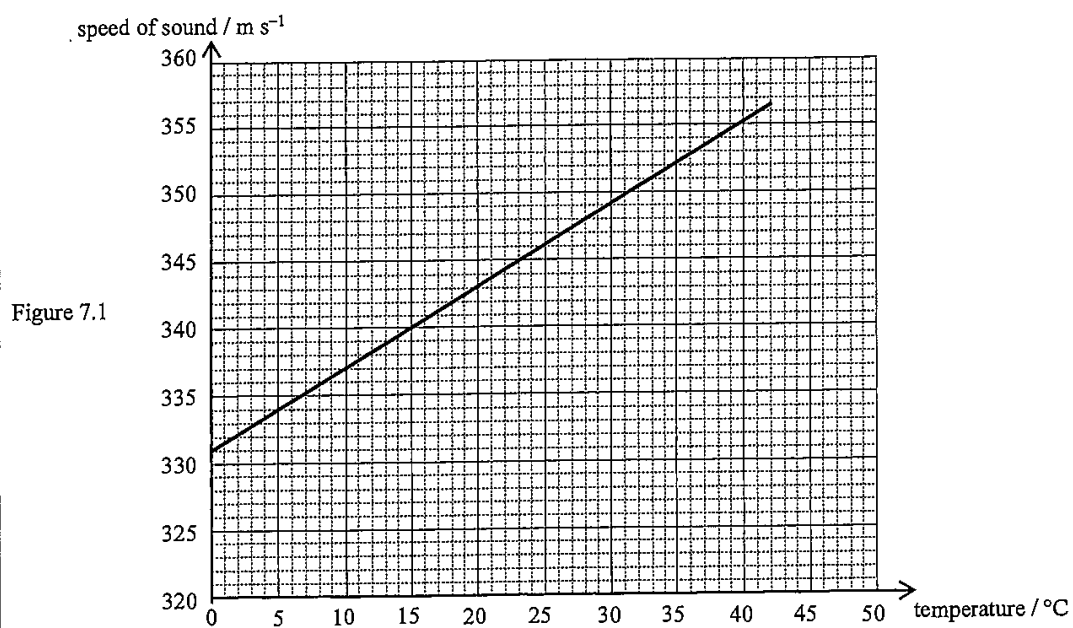
(3 marks)

(b) State and explain the change in x if green laser light is used.

(2 marks)

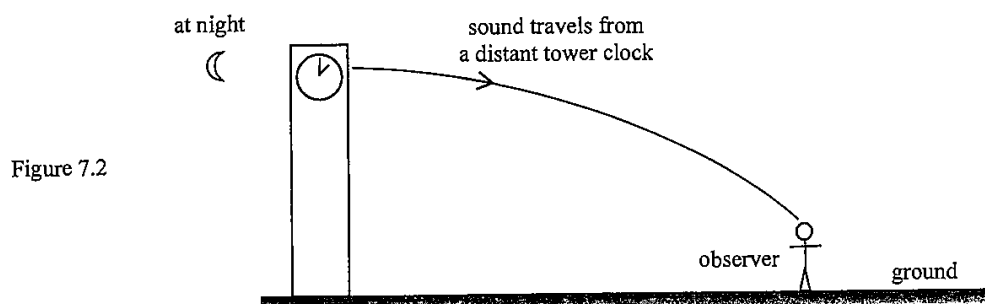
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7. The graph below shows the speed of sound at different air temperatures.



- (a) With reference to the graph, state the relationship between the speed of sound and the air temperature. (1 mark)

- (b) Figure 7.2 shows sound travelling from a distant tower clock to an observer at night when the ground cools.

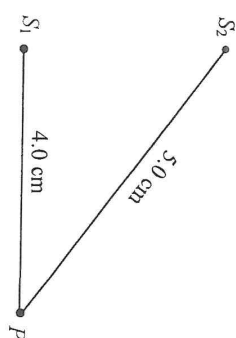


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(b) (i)

Figure 5.2



P is a point at the water surface as shown in Figure 5.2. Express the path difference at P from S_1 and S_2 in terms of λ .

(2 marks)

(ii) Explain briefly what type of interference occurs at P .

(2 marks)

(c) If the depth of the water in the ripple tank is increased while other conditions remain unchanged, deduce how the wavelength of the waves will be affected. (2 marks)

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